

FRED HOWARD PARK, FL, USA

WMO: 994008

Lat: 28.1533N

Lon: 82.8012W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.8	(c) 7.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.0	(k) 13.9	(l) 12.4	(m) 13.4	(n) 5.4	(o) 60	(p) 0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.7	(c) 31.9	(d) N/A	(e) 31.4	(f) N/A	(g) 30.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.2	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.3	(b) 9.9	(c) 8.6	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	15.4	17.1	18.8	22.2	25.1	27.7	28.0	28.4	27.2	24.2	19.6	17.6		
	DBStd	5.33	4.19	3.79	3.30	2.24	1.73	1.30	1.19	1.12	1.22	2.41	3.41	4.33		
	HDD10.0	13	8	1	1	0	0	0	0	0	0	0	0	3		
	HDD18.3	302	108	61	36	3	0	0	0	0	0	1	26	66		
	CDD10.0	4628	173	202	273	367	469	529	558	571	517	440	288	241		
	CDD18.3	1876	16	28	51	120	211	279	299	312	267	183	64	45		
	CDH23.3	17203	21	36	77	504	1708	3219	3505	3829	2756	1278	192	78		
(14) Wind	WSAvg	4.4	5.0	4.5	5.1	4.8	4.5	3.9	3.7	4.0	4.0	4.5	4.8	4.7		
	PrecAvg	1346	76	83	98	48	71	156	201	213	182	74	53	78		
	PrecMax	2037	202	277	374	124	385	465	565	470	460	359	232	396		
	PrecMin	931	2	8	0	0	0	16	24	65	11	1	0	0		
	PrecStd	241	42	60	79	37	75	92	100	82	98	72	48	79		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	26.2	26.6	28.5	30.7	32.5	32.7	32.9	32.4	30.6	28.3	26.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	23.6	24.2	25.1	27.4	29.7	31.6	31.6	31.9	31.5	29.3	26.7	25.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	22.5	23.3	24.3	26.6	28.9	30.8	31.0	31.4	30.7	28.4	25.5	24.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	21.3	22.2	23.5	25.8	28.2	30.3	30.5	30.8	30.1	27.8	24.3	23.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		6.4	5.7	5.9	5.4	4.9	4.6	4.6	4.7	5.4	5.7	6.3	6.1		
	5% DB	MCDBR	7.2	5.9	6.6	6.4	6.4	5.4	5.1	5.1	6.4	6.2	6.9	6.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.345	0.357	0.368	0.392	0.425	0.451	0.476	0.459	0.423	0.390	0.354	0.343		
	taud		2.523	2.495	2.454	2.354	2.319	2.322	2.261	2.323	2.391	2.466	2.511	2.553		
	Ebn at Noon		899	915	923	904	870	842	820	834	857	868	880	885		
	Edh at Noon		90	100	110	126	131	130	137	128	116	102	91	84		
(44) All-Sky Solar Radiation	RadAvg		3.50	4.21	5.38	6.48	6.91	6.26	5.90	5.67	5.15	4.74	3.84	3.22		
	RadStd		0.25	0.30	0.48	0.34	0.38	0.49	0.35	0.35	0.37	0.30	0.22	0.27		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	-4	-49	+167	+116

Nomenclature: See separate page